MIL-F-50415A (MU)
13 December 1973
SUPERSEDING
MIL-F-50415 (MU)
30 APRIL 1969

MILITARY SPECIFICATION

FUZE, ROCKET, M412A1 LOADING AND ASSEMBLING

1. SCOPE

1.1 This specification covers special requirements not normally required by the drawings and provides quality assurance provisions for the loading and assembling for one type of fuze designated as Fuze, Rocket, M412A1 for use in Rocket, HE, 66 Millimeter, Anti-Tank, M72A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-F-50407A - Fuze, Rocket, M412A1, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1340

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DRAWINGS

PICATINNY ARSENAL

8796377 - Pellet, Booster
9223224 - Fuze, Rocket, M412A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Fuze. - The fuze shall comply with all requirements specified on Drawing (dwg.) 9223224 and with all requirements specified in applicable specifications.

3.3 Workmanship. - All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.4 Position of Locking Shaft. - The locking shaft shall bear on the graze lever as illustrated in dwg. 9243295 as evidenced by 0.350 inches minimum distance between the inside bottom surface of the firing pin and the bottom surface of the mounting base when the firing pin is inserted into the fuze cavity of the mounting base.

3.5 Electrical Resistance. - The maximum electrical resistance, indicative of an unarmed fuze, shall not exceed 10 ohms when measured between the telltale contact in the cover assembly and the ground terminal and shall not be less than one megohm when measured between the ground terminal and insulated terminal in the base connector assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all materials purchased by the contractor meet requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required by subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 14 complete fuze assemblies and 45 booster pellets (dwg. 8796377) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (see 4.1.1.c).

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A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Each fuze, subassembly and component shall be examined and inspected for every dimension and requirement applicable to the fuze, subassemblies or components. Also, the entire sample shall be examined and tested in accordance with 4.3 as applicable (without regard to AQL or sample size specified therein).

4.2.3 Rejection. - The first article sample shall be rejected if one or more fuzes, subassemblies or components fail to conform to any or all of the applicable requirements. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with

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requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Fuze assemblies (less booster pellet) - (dwg. 9243295) from not more than one fuze assembly lot from one manufacturer.
b. Booster pellets from not more than one lot from one manufacturer.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Pellet, Booster (see dwg. 8796377 covering a detail of dwg. 9223224).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.25 Percent		
101.	Length, minimum (min.)	Gage	01001
102.	Cracked or chipped	Visual	01002
Minor: None defined.			

4.3.2.2 Fuze, Rocket: M412A1 (see dwg. 9223224).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor in armed or partially armed position	Visual (see 4.4.3)	02001

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2.	Rotor in armed or partially armed position (see 3.5)	Test (see 4.4.3)	02002
Major:	AQL 0.65 Percent		
101.	Length of cover plate, maximum (max.)	Gage	02003
102.	Total length, max.	Gage	02004
103.	Diameter at cup end, max.	Gage	02005
104.	Cover assembly loose	Manual	02006
105.	Component damaged or missing	Visual	02007
106.	Booster cup not crimped 360 degrees	Visual	02008
Minor:	AQL 1.00 Percent		
201.	Marking incorrect, incomplete or illegible	Visual	02009
202.	Evidence of poor workmanship (see 3.3)	Visual	02010

4.3.3 Testing

4.3.3.1 First Article Sample (see 4.2.1). - The samples submitted as the first article quantity shall be tested as specified herein after being examined 100 percent for the applicable classifications of defects in Paragraph 4.3.2.

4.3.3.1.1 Specific Gravity of Booster Pellet (see dwg. 8796377) - Major Defect - Code No. 01003. - A sample of 45 booster pellets shall be selected for the first article and tested as specified in 4.4.1 using test equipment specified in 4.3.4. Rejection shall be as specified in 4.2.3.

4.3.3.1.2 Position of Locking Shaft (see 3.4) - Major Defect - Code No. 02011. - In the event that the graze functioning elements of the fuze (less booster pellet) are received at the loading plant unassembled or if the graze elements have for any reason been removed, the following inspection shall be performed 100 percent. The inspection shall be performed following the insertion of the firing pin into the firing pin cavity and immediately prior to the insertion of the firing pin spring into the firing pin and the assembly of the base plug into the firing pin cavity. The inspection shall consist of gaging to determine if the fuze meets the applicable requirements. Fuzes failing to meet the requirement as determined using the inspection equipment specified in 4.3.4 shall be classed defective. Rejection shall be as specified in 4.2.3.

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4.3.3.2 Production

4.3.3.2.1 Specific Gravity of Booster Pellet (see dwg. 8796377) - Major Defect - Code No. 01003. - A sample of 45 booster pellets shall be selected from each lot for this test. The test shall be performed as specified in 4.4.1 using test equipment specified in 4.3.4. If any booster pellet fails to comply with the specified drawing requirement, the lot shall be rejected.

4.3.3.2.2 Position of Locking Shaft (see 3.4) - Major Defect - Code No. 02011. - In the event that the graze functioning elements of the fuze (less booster pellet) are received at the loading plant unassembled or if the graze elements have for any reason been removed, the following inspection shall be performed 100 percent. The inspection shall be performed following the insertion of the firing pin into the firing pin cavity and prior to the insertion of the firing pin spring into the firing pin and the assembly of the base plug into the firing pin cavity. The inspection shall consist of gaging to determine if the fuze meets the applicable requirements. Fuzes failing to meet the requirement as determined using the inspection equipment specified in 4.3.4 shall be classed defective and removed from the lot subject to repositioning of the locking shaft and re-inspection prior to re-entry into the lot.

4.3.3.2.3 Drop Test of Fuze, Less Booster Pellet - Fuze functions during test or is armed after test - Critical Defect - Code No. 02012. - Prior to assembly of the booster pellet, each fuze assembly (dwg. 9243295) shall be tested for partial functioning of the setback assembly in accordance with 4.4.2 using test equipment specified in 4.3.4. The test shall be performed 100 percent and fuzes classed defective shall be destroyed.

4.3.4 Inspection Equipment (see 6.2 and 6.3). - The inspection equipment required to perform the examinations and tests prescribed in 4.3.2 is identified under the "Method of Inspection" headings. Unless otherwise specified, the inspection equipment used to perform the examinations and tests prescribed in 4.3.3 shall be contractor designed (see 6.3.2).

4.4 Test Methods and Procedures

4.4.1 Specific Gravity of Booster Pellet. - The booster pellet shall be accurately weighed. Displacement of clean mercury by the booster pellet shall be determined and the displaced mercury shall be weighed. Calculate the specific gravity of the pellet as follows:

$$\text{Specific Gravity} = \frac{13.54W}{\text{the weight of the displaced mercury in grams}}$$

where:

W = Weight of the booster pellet in grams.

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4.4.2 Fuze Drop Test. - The assembly shall be positioned in the test fixture (drop tower) in a vertical, base down orientation. The assembly, when released, shall be subjected to free fall for a minimum height of forty (40) inches with impact on a hard, metallic surface. The assembly shall then be repositioned in a vertical, base up orientation and when released, shall be subjected to free fall for a minimum height of six (6) inches with impact on a hard, metallic surface. The assembly shall be classed defective if it functions explosively during performance of the test or if it is armed at the conclusion of the test.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.3 Fuze Safety. - The unarmed position of the rotor shaft shall be visually confirmed (see dwg. 9243295). Any fuze containing an armed or partially armed rotor shall be classed defective and removed from the lot. After determining that the fuze is visually safe, the electrical resistance shall be measured between the following points: between the telltale rivet (10048792) and the connector pin (9251973) and between the straight connector pin (9251969) and the connector pin (9251973). The fuze shall be classed defective and removed from the lot if it fails to meet either electrical requirement.

WARNING: When measuring the resistances, the test equipment shall limit the current to 20 microamperes to prevent inadvertent functioning of the detonator. This test may be dangerous and should be performed behind a suitable barricade.

5. PREPARATION FOR DELIVERY

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection Code Numbers. - The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. These code

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numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.2).

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-QA-I.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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Preparing Activity:
Army-MU

Project Number: 1340-A-397

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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